

Aaryaman Arora

Biotechnology undergraduate with hands-on experience in web development and data-driven projects, and a strong interest in interdisciplinary research at the interface of biotechnology and biomedical engineering. Passionate about AI-driven biomedical systems and product -focused problem solving.

✉ aaryaman1407@gmail.com

☎ +91 9041092164

📍 India

🌐 aaryaman.me

🔗 /Aaryaman-147

📄 /in/aaryaman-arora

PROJECTS

HEALTHTECH

RESEARCH

IOT

Smart Wound Monitoring – Proof of Concept

Arduino · Sensors · Biomedical Instrumentation · Embedded Systems · Research Prototype

Proposed and prototyped a non-invasive smart bandage system for early graft failure detection using embedded sensors and data visualization.

MACHINE LEARNING

AI INFRASTRUCTURE

BioCUDA

PyTorch · NVIDIA Triton (Flash Attention) · CUDA Graphs · FastAPI · React.js · bitsandbytes (INT8 Quantization) · Scikit-Learn

A dual-branch, high-throughput machine learning infrastructure for executing 3 Billion parameter foundational models (ESM-2) on constrained hardware and enterprise A100 clusters.

MLOPS

CLOUD INFRASTRUCTURE

End-to-End MLOps Pipeline

Python · ML Pipelines · Automation · Model Deployment · Docker · Apache Airflow

Designed an end-to-end MLOps pipeline for model training, versioning, and deployment. Automated data preprocessing, model evaluation, and deployment workflows

PRODUCTIVITY

TOOLING

Second Look – Browser Extension

JavaScript · Browser APIs · Extensions · UX

Developed a browser extension to improve user workflows and efficiency

PRODUCTIVITY

TIME MANAGEMENT

Schedly – Timetable & Scheduling App for University Students

Frontend · Logic Design · UX · JavaScript

Designed and built a timetable management application tailored for university students to organize classes, labs, and academic commitments.

GRAPHICS

CREATIVE CODING

Interactive 3D Particle System

JavaScript · WebGL / Three.js · Animation · Performance

Built a real-time interactive 3D particle system for web environments

EDUCATION

Thapar Institute of Engineering and Technology

B.Tech | Biotechnology

2024 - 2028 (Expected)

Innocent Hearts School

Senior Secondary (Class XII & X)

2019 - 2024

PCM

Cambridge International School

Foundation & Middle School

2013 - 2019

Early education and foundational studies.

LEADERSHIP & VOLUNTEERING

Executive Committee Member – Research & Development

Nucleus

2026 - Present

Core Member – Logistics

Saturnalia

2025 - Present

Managed on-ground execution for a large-scale college festival. Coordinated resource allocation and crowd flow across multiple teams.

Volunteer

PFA (People For Animals)

2025 - Present

Active member of the university chapter dedicated to animal welfare and rescue operations.

TECHNICAL SKILLS

Languages

JavaScript (ES6+)

Python

AI/ML

HEALTHCARE AI

AI Health Risk Stratification Assistant

React (Next.js) · Tailwind CSS · Python (FastAPI) · Node.js (API integration) · LLM APIs · OpenRouter (multi-model routing) · Retrieval-Augmented Generation (RAG)

Built a hybrid AI-based health risk assessment system using LLMs, RAG, and rule-based safety guardrails to generate structured, explainable triage outputs.

ACHIEVEMENTS & HACKATHONS

Hack4Help (Israeli Indian Hackathon)

Top 10 Finalist | Team Lead

Led a multidisciplinary team to design a smart bandage system. Qualified through multiple screening rounds to present a functional hardware-software demo in the finals.

- Integrated temperature sensors with Arduino for real-time monitoring.
- Oversaw development of a live data dashboard and coordinated integration of hardware and software components
- Presented the final in-person pitch and functional demo during the 24-hour hackathon
- Selected among Top 10 teams based on innovation and feasibility.

Spark Tank Pitch – IIT Delhi

Presenter

Pitched an advanced smart bandage concept involving multiple biomarkers. Presented technical architecture and clinical relevance to an expert jury. Received feedback on translational potential and scalability in healthcare settings

EXPERIENCE

Independent Developer

2024 - Present

Designed and built multiple web applications and tools focusing on real-world use cases. Worked with APIs, frontend logic, and UI design to deliver functional, user-focused products

Frameworks

React.js

Three.js

Node.js

Tailwind CSS

Flutter

Tools

Git

GitHub

Figma

VS Code

Chrome DevTools

Vercel

Soft Skills

Leadership

Team Collaboration

Problem Solving